Decoding 10,000 years of ecological persistence through causal networks

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ABSTRACT

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1 Introduction

The persistence and reorganisation of ecological communities over long timescales represent a fundamental challenge in modern ecology. As anthropogenic pressures push the Earth system beyond its safe operating boundaries, identifying the mechanisms that enable communities to maintain functional integrity despite severe disturbances has become an urgent priority. While contemporary short-term studies offer critical insights into species interactions, they frequently overlook the slow dynamics, feedback loops, and legacy effects that define ecosystems over centuries and millennia. To formulate robust ecological forecasts, it is essential to trace the baseline conditions of ecosystem complexity and stability through long-term paleoecological records.

Modern coexistence theory provides a rigorous framework for understanding these dynamics by distinguishing between niche-based processes (interaction assembly) and stochastic factors (neutral assembly). In the context of resilience, stable coexistence is underpinned by the balance of stabilising mechanisms—such as niche partitioning—and fitness differences. A community's resilience is often reflected in its network architecture; for instance, high levels of nestedness or modularity can influence how a system recovers from perturbation. When communities are governed by niche partitioning, the resulting functional independence allows species to respond to stressors in a non-synchronised manner, thereby enhancing the overall robustness of the ecological network. Conversely, a shift toward neutral assembly or extreme environmental filtering can simplify these networks, potentially reducing the community's capacity to withstand critical transitions. Fire activity has proven to be a primary modulating force in these assembly processes, functioning as a selective pressure that can radically reorganise causal dependencies between taxa. As a major stochastic disturbance, fire regimes may determine predictability relationships within a community.

In this study, we move beyond individualistic species-response models to examine how whole-community properties emerge and persist through time. We focus on high-altitude subalpine and afroalpine ecosystems, as mountain regions serve as critical natural laboratories; they are biodiversity hotspots that have acted as floral refugia during the Quaternary but are currently among the environments most endangered by global warming and land-use changes.

By applying causal inference—specifically Granger Causality—to continuous Holocene pollen and charcoal records (ca. 10,000 years), we reconstruct time-varying networks to determine how environmental filtering and fire activity have modulated community assembly patterns.

We found that in these regions under high-intensity fire regimes, the prevalence of positive causal links suggests a synchronised community response; taxa share both vulnerabilities and resources, responding to the disturbance as a cohesive whole. In contrast, periods of lower fire activity are characterised by a higher proportion of negative links. This shift indicates that in the absence of severe environmental forcing, more subtle interactions—such as competition, facilitation, and resource distribution—take precedence, allowing for functional independence and the establishment of stable, diverse communities. Understanding how these fire-driven shifts impact network complexity over millennial scales is therefore vital for assessing long-term ecosystem stability.

This approach allows us to quantify the transient dynamics of ecological networks, providing a vital bridge between neo- and palaeoecological sciences.

2 Results

@@@ COSAS A COMENTAR: @@@@

1. CREATION OF CAUSALITY NETWORKS (TABLE WITH BASIC NETWORK FEATURES)
2. ABUNDANCE OF NEGATIVE LINKS IN SNAPSHOTS WITH ANOMALOUS FIRE ACTIVITY (HISTOGRAMS WITH FRACTION OF NEGATIVE EDGES)+ Taxa dynamics and degree distribution. Out-degree distributions revealed shifts in species influence:
   (a) In the more active fire-prone periods, a few species exerted disproportionate influence.
   (b) During the low-fire periods, influence was more evenly distributed.
3. COARSE GRAINED VERSION OF CAUSALITY NETWORKS (FIGURES WITH NETWORKS)

2.1 Time series of taxa at the extraction sites

We work with the sedimentary samples extracted from two sites: Basa de la Mora (BSM) and Garba Guracha (GGU) (see Methods). As the original sedimentary samples are unevenly spaced in time, taxa abundances records have been converted into evenly spaced time series via a Generalized Additive Model (GAM) **@@ CITA GAM @@**, allowing for robust time-series analysis (see Methods). Figure 1 displays the processed abundance time series for the ten most abundant taxa in both sites. Furthermore, to investigate how environmental perturbations affect ecological community structure, we divided the time series into pieces (snapshots) based on historical records of charcoal concentrations (see Methods and Supplementary Figure 1). [falta añadir las ventanas en la figura 1]. These snapshots encapsulate regimes of high, low, or intermediate fire intensity which may drive the system's dynamics.

When interpreting the data, it is important to keep in mind that the pollen counts are used as a proxy for plant biomass or absolute population size but, they do not represent a direct measurement of either quantities. Indeed, certain species produce more pollen than others, and variations in pollen capacity to travel over long distances introduces representational biases in the total abundance computed for a specific taxa. That said, we want to remark that the outcome of the causal inference does not depend on the overall scale of the time series, as the results of the vector autoregression do not change when the values of the abundances are normalized [referencia?].

@@ ESTA FRASE LA DEJO PERO SEGURAMENTE HABRÁ QUE ADAPTARLA @@ Again, as observable in Figure 1, the pollen abundances do not mirror the evolution of plant communities and the effects of fire intensity beyond general trends. To overcome this limitation, we turn to causal inference.

2.2 Topological properties of the causality networks

To represent the directional interdependencies between taxa going beyond their mere co-occurrence, we modeled the ecological interactions using causal inference. In observational ecology, several inference methods exist to serve this purpose, like: Structural Equation Modelling (SEM), Empirical Dynamic Modeling (EDM), Bayesian networks, Transfer Entropy (TE), and Granger Causality (GC) **@@ CITA REVIEW DE INFERENCIA Y SISTEMAS ECOLOGICOS Y/O OTRAS FUENTES @@**.

Although these methods have differences **@@ SI TENEMOS UNA REFERENCIA ES EL CASO DE PONERLA @@**, the majority of them rely on common assumptions to be accounted for. Most notably, they require us to assume that there are no unknown or omitted confounding variables, meaning that all the relevant variables that could translate into biased results have been measured. Some of these methods, such as SEM, also require specifying a plausible causal structure. These models often make assumptions about the data's properties **@@ LIKE WHAT? @@**. Although EDM methods can handle non-linear relationships without assuming separation of variables, they are meant to deal with highly deterministic systems, which are not

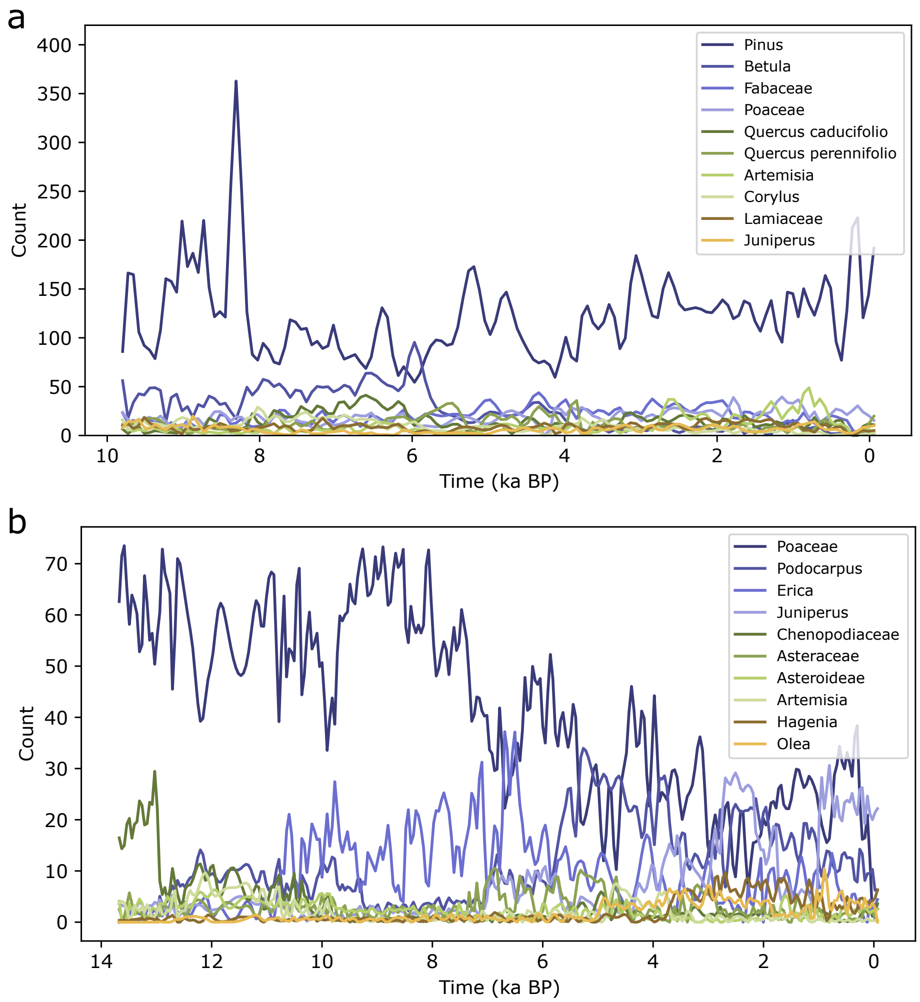

Figure 1. Ten taxa with the most abundant pollen production for BSM (panel a) and GGU (panel b) sites. For each site, we display the continuous time series of count in pollen grains as a function of time (in ka BP). Original data were smoothed using GAM (see Methods).

the norm for systems spanning thousands of years. Furthermore, EDM and TE methods also require a very large number of measurements to reconstruct the intrinsic dynamics accurately.

Given the size limitations of the datasets, and lack of prior assumptions regarding the underlying causal structure of the system, we chose the Granger Causality for our analysis[1]. Although GC relies on assumptions like linearity, separability of variables, and lack of confounding variables; we consider it more suitable than other methods (*e.g.*, EDM or TE) given its reduced number of degrees of freedom. That said, as GC makes assumptions about the data which strongly skew the results, we adopted stationary bootstrapping[2] to ensure the statistical robustness of the results (see Methods). Then, we generate *causality networks* of taxa to map the causal relationships existing within our systems.

Networks provide a simple, yet powerful, paradigm to grasp the structure and dynamics of the relationships of a wide variety of complex systems[3,4]. The versatility of network science has led its application to many disciplines, from social sciences[5] to biology, from neuroscience[6] to physical systems[7], to cite a few, including ecology[8].

Leveraging such a framework, we model the ecological interactions between taxa found through GC as a *directed*, *weighted*, and *signed* network[3]. In these networks a node represents a taxon, and a link between taxa denotes a causal relationship between them (see Methods). We considered 79 nodes for BSM and 83 for GGU which, in turn, translated into between 190 and 253 links for the snapshots of BSM, and between 91 and 308 links for those of GGU, instead. Table 1 (see Methods) summarizes the main features of the snapshots (*i.e.*, networks) obtained for each time interval considered for BSM and GGU, respectively.

To give more sense to the web of interactions, and map them onto known ecological communities, we grouped the taxa into *functional groups* designated by an expert (see Tab. [TABLA EN MATERIAL SUPLEMENTARIO] for the details). Figure

2 portrays such a coarse-grained version of the causality networks. A quick glance at Fig. 2 highlights that the network's wiring evolves considerably over time. **@@ PROPONGO CALCULAR LA TEMPORALIDAD NO ENTRE ESPECIES SINO ENTRE GRUPOS FUNCIONALES. CREO QUE VA A SALIR MEJOR QUE AQUELLA ENTRE ESPECIES, Y NOS PERMITE DAR UN TONO MAS CUANTITATIVO A LA DESCRIPCIÓN DEL PATRON DE CONNEXIONES ENTRE NODOS @@**

Another feature of our networks is the abundance of positive and negative links. **@@ AQUI TENEMOS QUE PONER UNA FRASE DE CONEXION CON EL PRÓXIMO PARRAFO @@**

2.3 Influence of fire activity on ecological interactions

We divided the original time series according to the regimes of fire activity, and we have found that, as expected, fire events exert a strong effect on the organization and nature of the causal relationship existing between taxa. More specifically, the proportion of negative causal links f_- – *i.e.*, the fraction of total amount of links, E, for which an increase of pollen abundance in a taxon's past, results in a decrease in the abundance of another taxon, – varies significantly over time, hinting a the existence of some correlation with fire anomalies. As summarized in Tab. 1 and Fig. 3, periods characterized by elevated fire intensity exhibit smaller values of f_-. Conversely, periods of lower fire activity display higher percentages of negative relationships (*i.e.*, low values of f_-).

The prevalence of positive links during periods of elevated fire intensity suggests the existence of tightly intertwined dynamics among taxa spurred by strong external disturbances. Under frequent environmental changes, taxa tend to be affected in similar ways, improving their predictive capabilities (albeit some of them decayed or recovered quicker than others). Within these harsh regimes, taxa are more likely to respond to the disturbances as a whole rather than interfering with each other, as they share both resources and vulnerabilities, thus resulting in an increase of both the positive, $1-f_-$, and overall number of links, E. On the contrary, the higher fraction of negative links in lower fire intensity snapshots can be interpreted as a sign of functional independence and niche partitioning. In mild disturbance periods, the lack of strong stochastic environmental effects gives room to more subtle types of interactions to govern the overall dynamics – such as competition, facilitation, and resource distribution. Such a change of regime allows species to compete for distinct functional roles and, therefore, results in larger amounts of negative interactions.

These results contribute to the understanding of organization and resilience of ecological communities[9]. These stable periods provide a way for species to find their niche and build stable communities. [CITAS (GODOY), DISCUSION SOBRE RESILIENCIA]

[DISCUSION DE LA DISTRIBUCION DE GRADOS CON LOS REGÍMENES DE FUEGO]

3 Methods

3.1 Nature of the data and studied sites

Plant community composition was reconstructed using fossil palynological assemblages, while fire dynamics were inferred from sedimentary macro-charcoal particles. Our analyses drew upon continuous, high-resolution sedimentary sequences previously retrieved from two geographically distinct natural laboratories: Basa de la Mora (BSM) in the Central Pyrenees, Spain (1913 m a.s.l.), and Garba Guracha (GGU) in the Bale Mountains, Ethiopia (3950 m a.s.l.).

Taxonomic resolution for fossil pollen is typically constrained to the genus or family level; thus, we classified pollen assemblages into functional plant communities based on established regional floristic frameworks: Villar et al. (1997, 2001) for the Pyrenean record and Miehe & Miehe (1994) for the afromontane sequence. Aquatic taxa were excluded from the final datasets as their dynamics are primarily governed by local limnological conditions rather than the broader upland community assembly processes we aimed to quantify.

3.1.1 Basa de la Mora (BSM)

Lake Basa de la Mora is a proglacial basin situated in the Cotiella Massif (Central Pyrenees, Spain). The site occupies a critical ecotonal position influenced by Mediterranean climatic patterns, characterized by seasonal precipitation peaks and a mean temperature range of 0.5°C to 15°C. The surrounding current vegetation exhibits a complex mosaic of altitudinal zonation:

- Lowlands and Montane Belt: Dominated by riparian forests (Fraxinus excelsior, Populus spp.) and mixed montane woodlands featuring Pinus sylvestris, Fagus sylvatica, and Quercus faginea.
- Subalpine and Alpine Zones: Characterized by the retreat of the treeline, where Pinus uncinata forests transition into Juniperus communis scrublands and hardy alpine grasslands.

The BSM record spans approximately 9,8 ka PB (calibrated kilo years Before Present), encompassing nearly the entire Holocene, with 140 samples providing decadal to centennial resolution[10].

@@@ PONER AQUI EL LISTADO DE TAXA AGRUPADAS POR COMUNIDADES @@@@

3.1.2 Garba Guracha (GGU)

Garba Guracha is a cirque lake located at 3950 m a.s.l. on the Sanetti Plateau of the Bale Mountains National Park, Ethiopia. The site is strategically positioned at the transition between the Ericaceous and afroalpine vegetation belts[11].

- Afroalpine and Ericaceous Belts: Local communities are dominated by Alchemilla haumannii and Helichrysum shrublands, transitioning into Erica trimera and E. arborea thickets.
- Afromontane Forest Limit: The lower slopes feature dry afromontane forests dominated by Juniperus procera, Hagenia abyssinica, and Podocarpus falcatus.

This 15.5 -metre sedimentary sequence provides a 14 ka BP record of high-altitude tropical vegetation dynamics.The GGU dataset comprises 301 pollen samples and 1301 charcoal samples, allowing for a detailed examination of fire-vegetation feedbacks.

@@@ PONER AQUI EL LISTADO DE TAXA AGRUPADAS POR COMUNIDADES @@@@

3.1.3 Data metrics and processing

Palynological research frequently relies on percentages (relative abundance) estimated as the fraction of a particular taxon over the total counts of pollen grains on microscope slide. This metric is fundamentally unsuitable for causal inference due to the mathematical closure effect. In the context of Granger Causality (GC), which identifies directional predictability by testing if the history of Taxon X improves the forecast of Taxon Y, using percentages introduces severe representational biases. Under these conditions, a negative causal link may represent a mere mathematical artifact of the closure effect rather than a genuine ecological interaction, such as competition or niche partitioning. To move beyond these constraints and achieve a more accurate proxy for absolute plant biomass and population size, we advocate for the prioritisation of Pollen Accumulation Rates (PAR). PAR estimates the absolute rate at which pollen is deposited per unit area per year (grains · cm^{-2} · yr^{-1}), calculated as follows:

PAR(i)= grain count · (counted Lycopodium spores/added Lycopodium spores) · sedimentation rate/ · sample volume. Lycopodium spores are used as an external marker and the archive specific sedimentation rate is needed to derive a multiplier, α, which allows each taxon's 'pollen rain' to be tracked independently. PAR serves as a reliable proxy for absolute plant biomass and population size, allowing each taxon to be tracked as an independent variable. This is achieved by incorporating sedimentation rates and an external marker (Lycopodium spores) to calculate a multiplier that isolates the 'pollen rain' from relative fluctuations.

We split the time series into several adjacent non-overlapping subsets corresponding to distinct regimes of fire activity (see Supplementary Figure 1), quantified according to charcoal concentration[10]. Such a procedure leads to consider three periods for the BSM site: i) from 9.8 to 6.2 ka BP (high fire activity), ii) from 6.2 to 3.8 ka BP (low fire activity), and iii) from 3.8 ka BP to present time (intermediate fire activity). The same criterion leads, instead, to identify four periods for the GGU site, namely: i) from 13.6 to 11.1 ka BP (low fire activity), ii) from 11.1 to 6.8 ka BP (high fire activity), iii) from 6.8 to 2.2 ka BP (low fire activity), and iv) from 2.2 ka BP to the present (high fire activity). The sediment samples have been extracted in a contiguous and equally spaced manner. Then, they have been dated and analyzed to obtain the grain counts (*i.e.*, number of pollen grains from each taxon in the sample), sedimentation rate, the sample's volume, and the concentration of *lycopodium* spores (both the amount added in the sample, as well as counted during the analysis)[12]. These information allow for the estimation of the so-called pollen accumulation rates (PAR) in every sample, i, as follows:

$$\mathrm{PAR}(i) = \text{grain count} \cdot \underbrace{\frac{\text{added lycopodium spores}}{\text{counted lycopodium spores}} \cdot \frac{\text{sedimentation rate}}{\text{sample volume}}}_{multiplier} \equiv PC(i)\,\alpha(i)\,. \tag{1}$$

The *multipliers*, α, allow to find the rate at which pollen from all taxa accumulated over time in addition to the relative abundances, $PC(i)$, (*i.e.*, a taxon's, i, pollen grain count over the total grain count of all taxa), allowing to use the PAR time series as the *causing* or *caused* variables. This reasoning applies to the the GGU sequence. For the BSM site, however, the lycopodium count is unavailable, and only the relative abundances are considered, instead. As conditional variables included when carrying out the inference, we considered the multipliers, $\{\alpha\}$, in both sites and solely sedimentation rate and sample volume for BSM. The inclusion of conditional variables accounts for potential biases introduced by environmental variations or by measurements. Section **??** of the SM presents a deeper analysis of the choice of variables (PC versus PAR) and the impact of conditional variables (*e.g.*, the multipliers). **@@@ FALTA HABLAR DE LA TEMPERATURA (DELTA 13) COMO CONDICIONAL @@@** añadir delta13

The sedimentary samples are analyzed homogeneously in depth, translating into a non-homogeneous sampling over time (*i.e.*, depth does not translate directly into time as sedimentation is not uniform). Such a dis-homogeneity requires that our original time series must undergo some kind of interpolation process to convert them into evenly spaced time-series. The latter step is required to use the machinery of time-series analysis, which assumes that time-series are evenly sampled. Our original time series comprised a total of 140 samples in BSM and 301 in GGU which have been interpolated using a Generalized Additive Model (GAM)[13,14] to resample them. GAM uses splines to fit our predictor, the pollen abundance, and adds a penalty for smoothness ensuring the outcome is not a jagged line. It is implemented using the `GAM` function from the Python library `pyGAM`[14], **@@ NO ENTIENDO MUY BIEN ESTA FRASE @@** and including the abundances over time as a *spline* term. GAM allows us to tune its parameters to avoid underfitting, effectively capturing the short-term fluctuations of our data. The number of splines is set to the number of samples minus one, and the smoothing parameter, λ, is set to 0.01, ensuring the preservation of the perturbations which are essential to carry out statistical inference.

To ensure statistical robustness, we excluded from the analysis taxa with extremely low pollen counts (*i.e.*, removing, in every time period, those time series with less than three values higher than 0.5); as their GAM interpolations correspond to close-to-zero curves, which can be assimilated to noise. Additionally, aquatic species have been omitted from the analysis to limit the scope to spatially co-occurring communities.

3.2 Inferring causality

3.2.1 Granger causality

Spatial or temporal co-occurrence or co-abundance[15] does not always imply the existence of an interaction between species[16]. Causal inference is a branch of statistics aiming at determining the cause-and-effect relationship between variables, going beyond simple correlation.

The standard test used to infer causality (*e.g.*, in medical trials) is the Randomized Controlled Trials (RCTs), in which we are able to manipulate the environment to find a result **@@@ CITAR ALGO SOBRE RCT @@@**. However, in our case, we cannot use such a method as we are compelled to use observational data in the form of time series. Therefore, we resort to observational causal inference **@@ YO PONDRÍA DISCOVERY EN LUGAR DE INFERENCE @@**, which is widely used in areas such as economics and neuroscience **@@@ CITAR PAPER MARIO CHAVEZ Y ALGO DE ECONOMICAS @@@**. This framework does not prove causal relationships in the strict philosophical sense but, rather, tests for its *predictability*. In this sense, variable X is said to mathematically *cause* Y if knowing the past values of X improves the prediction of Y's future values better than knowing Y's past values alone.

Amidst the plethora of methods available to estimate causal relationships from time series[17], we choose the so-called Granger Causality (GC)[1] to test for causal relationships between pairs of taxa. The GC compares two Vector Autoregressive Models (VARs): the unrestricted (in which both the *caused* and *causing* taxon's past values are taken into account) and and restricted (in which only the *caused* taxon's past values taken into account).

Additionally, these VAR models allow to introduce external (*i.e.*, *conditional*) variables to account for confounding factors that could explain some of the observed causal links. Given two time series $X \equiv \{x_1, \ldots, x_T\}$ and $Y \equiv \{y_1, \ldots, y_T\}$, and indicating their values at time t as X_t and Y_t, the unrestricted Granger Causality model is written as:

$$Y_t = c_0 + \sum_{i=1}^{h} \alpha_i X_{t-i} + \sum_{j=1}^{h} \beta_j Y_{t-j} + \sum_{p} \sum_{k=1}^{h} \gamma_k^p C_{t-k}^p + u_t \ , \tag{2}$$

@@@ TENEMOS UN PROBLEMA, EL SIMBULO α YA SE HA USADO EN LA EQUACIÓN DEL PAR ... HAY QUE ENCONTRAR OTRO SIMBULO @@@

where c_0 is the intercept, β_i are the variable Y (*i.e.*, *caused*) own-lagged coefficients, α_i are the (*causing*) variable X cross-lagged coefficients, and γ_k^p are the cross-lagged coefficients of the *conditional* variables C^p **@@ CITA ALGUN TEXTO BASE SOBRE GRANGER CAUSALITY @@**. Additionally, u_t are the residuals, which are assumed to have a normal distribution **@@@ PABLO REVISA SI ES CORRECTO LO QUE DIGO @@@** (*i.e.*, a white noise). Lastly, h is the maximum time lag, which we set to one to avoid overfitting, given the limited number of data points. Under the assumption that $h = 1$, β_1 represents the autocorrelation component of the time series Y, whereas α_1 accounts for the causal effect that X has on Y. **@@ CUAL SON LOS VALORES POSIBLES DE GC? TENEMOS QUE EXPLICAR QUE QUIEREN DECIR LOS VALORES EXTREMOS (MAX NEGATIVO, CERO, MAX POSITIVO) @@@**

@@ PABLO, REVISA QUE ESTA FRASE ESTÉ BIEN @@ Confounding variables can influence the causal inference between two variables by simultaneously influencing both the presumed *causing* and caused time series, resulting in artificial relationships. For this reason, whenever possible, it is important to include information accounting (even partially) for shared drivers as conditional variables. For BSM, we have considered two conditional variables: the *multipliers*, $\{\alpha\}$, introduced in Eq.(1) (without the lycopodium information), and the amounts of cave isotope composites for δ13C found in Mendukilo[18], ~~which represents~~ **denoting** a reconstruction of temperature variability in the area. For GGU sequence, instead, we have included only the former (*i.e.*,, $\{\alpha\}$) as a conditional variable. A sensitivity analysis on the effects of the *multipliers* as conditional variable can be found in Section @@@@ of the Supplementary Material.

3.2.2 Bootstrapping

In principle, the residuals u_t can be used to compute the F-Statistic of the GC's values to estimate theoretically the p-value to determine whether a causal relationship between two variables exists (*i.e.*, it is statistically significant) or not. However, the necessary mathematical assumptions behind such an approach rarely hold true when dealing with empiric data. The use of *bootstrapping* is a valid alternative to test the causality test's null hypothesis, and estimate then the p-values for all pairs of variables **@@ CITAR ALGUNA REFERENCIA BASE SOBRE BOOTSTRAPPING @@**. In particular, bootstrapping addresses several issues: i) small sample sizes (as the F-Statistic relies on asymptotic theory and assumes infinite data points), ii) the assumption of normally distributed residuals, iii) the data's autocorrelation, and iv) the evolution over time in the variance of non-stationary series. Overall, the p-values estimated via bootstrapping are less skewed towards lower (*i.e.*, more statistically significant) values than the theoretical ones. Among the several bootstrapping methods available, we chose the so-called *stationary bootstrap*[2,19].

Stationary bootstrap resamples contiguous blocks of data from the original series of length, T, to generate a new series by pasting blocks of random lengths $t_i' < T$ until getting a series whose size is equal to the original (*i.e.*, $\sum_i t_i' = T' = T$). Stationary bootstrap, unlike the so-called independent and identically distributed (IDD) bootstrapping, has the advantage of preserving the data's autocorrelation **@@ AÑADIR CITA @@**.

For every pair of taxa in our datasets, we resampled the time series 1,000 times and obtained their respective causality coefficient α_1. Then, for each pair of taxa (i, j) the p-value associated with $\mathrm{GC}(i, j)$ is the proportion of bootstrapped tests for which the condition $\left|\mathrm{GC}'(i, j)\right| > \left|\mathrm{GC}(i, j)\right|$ holds (*i.e.*, the GC in the bootstrapped pair has a higher absolute value than the original).

3.3 Causality networks

We represent the causal relationships between taxa as a network (or graph), $G(\mathcal{N}, \mathcal{K})$[3]. In such a network, nodes are the taxa, and a link between taxa i and j, $(i, j) \in \mathcal{K}$ denotes a statistically significant causal relationship between them (*i.e.*, the p-value of $\mathrm{GC}(i, j)$ is less or equal to 0.05). Given the intrinsic nature of the GC, the links of G are directed, signed, and weighted; with the second corresponding to the sign of the GC (*i.e.*, $s(i, j) = \frac{\mathrm{GC}(i,j)}{|\mathrm{GC}(i,j)|}$), and the latter denoting its value (*i.e.*, $w(i, j) = |\mathrm{GC}(i, j)|$). We built a causality network for each time period in our datasets (*i.e.*, three for BSM and four for GGU). In the following, we describe several fundamental macroscopic measures of our networks.

For instance, the *degree* of a node is equal to the number of links incident with it. As links have a direction, the degree naturally splits into its *in*- and *out*- components. The former denotes the number of taxa Granger-causing the abundance of a taxon, whereas the latter indicates the number of taxa that a given taxon Granger-causes. In an ecological context, a taxon with a high *out*-degree can be thought as a proxy for the prediction of multiple taxa; either because it causes a variation (increase or decrease) in the abundance of other taxa or, because it responds faster to ecological events and perturbations (*e.g.*, wildfires)[20]. On the other hand, a taxon with a high *in*-degree can be thought as being caused by several taxa, or having a delayed response to external perturbations.

The size of the *Giant Connected Component* (GCC) of the network is defined as the number of nodes belonging to the biggest component, N_{GCC}, divided by the total number of nodes in the network, N. A component is a set of nodes connected by paths which can be reached from each other[3]. A value of GCC = 1 denotes that all nodes in the network belong to the biggest connected component, whereas a value of GCC = 0 corresponds to a network made of *isolated* nodes. Hence, GCC accounts for the *overall cohesion* of the system. In our case, we disregarded the links' directions when computing the size of the components.

Beside the network of causal relationships between taxa, one can study also its *coarse-grained* counterpart, *i.e.*, a new graph mapping the original network, $G(\mathcal{N}, \mathcal{K})$, (with N nodes and E links) into a smaller one, $G'(\mathcal{N}', \mathcal{K}')$, (such that $N' < N$ and $E' < E$). Several coarse-graining methods have been proposed (see, for instance,[21] and references therein) but, we decided to make a custom one. Specifically, in our coarse networks a node represent a community of taxa (*e.g.*, pastureland) and a link between two communities, a and b, $(a, b) \in \mathcal{K}'$ denotes the existence of links between taxa belonging to these communities. In agreement with the original network, G, the links of G' are also directed, signed, and weighted. Although the first two properties map directly between graphs, the weight of a link in G', w', does not indicate the value of the GC but, instead, the number of statistically significant links existing between taxa belonging to two communities.

4 Discussion

The reconstruction of ecological network dynamics over millennial timescales reveals that the persistence and reorganisation of plant communities are not merely responses to individualistic environmental tolerances, but are governed by complex,

Table 1. Topological properties of the inferred causality networks obtained from the Basa de la Mora and Garba Guracha sites across different time windows. For each site and each snapshot, we report its duration, the fire anomaly, the number of nodes (N) and of edges (E), the percentage of negative links (f_{-}), the relative size of the Giant Connected Component (GCC), and the average degree ($\langle k \rangle$).

Snapshot ID	Duration (ka BP)	Fire	N		E		$\langle k \rangle$		GCC		f_{-} (%)	
			original	coarse	original	coarse	original	coarse	original	coarse	original	coarse
Basa de la Mora												
1	9.8 – 6.2	High	79	8	253		3.20		70	1	23.3	
2	6.2 – 3.8	Low			190		2.41		69		40.5	
3	3.8 – Pres.	Medium			202		2.56		76		32.2	
Garba Guracha												
1	13.6 – 11.1	Low	83	8	91		1.09		45	1	37.4	
2	11.1 – 6.8	High			287		3.46		68		17.8	
3	6.8 – 2.2	Low			245		2.95		70		38.8	
4	2.2 – Pres.	High			308		3.71		81		23.1	

time-varying interdependencies. By applying causal discovery to high-resolution palynological records from two, high-altitude records, we reveal that a disturbance like fire may have acted as a modulator of network architecture.

Thus, a central finding of this study is the inverse relationship between fire intensity and the prevalence of negative causal links (f_{-}) within the ecological network. As illustrated in Figure 3, periods of lower fire activity — such as 6.2–3.9 ka BP in the Pyrenees and 6.8–2.2 ka BP in the Bale Mountains — are characterised by a significantly higher proportion of negative relationships, reaching up to 40.5% in BSM. In the framework of modern coexistence theory, these negative links represent a signature of functional independence and niche partitioning. In the absence of severe stochastic forcing, more subtle interactions such as competition and facilitation govern community dynamics, allowing species to occupy distinct functional roles and fostering high-altitude stability.

Conversely, periods of elevated fire activity (e.g., 9.8–6.2 ka BP in BSM) show a dramatic decline in negative links, with f_{-} dropping to approximately 9.9%. This prevalence of positive links suggests that taxa respond to severe disturbances as a synchronised whole. Under harsh regimes, shared vulnerabilities to the environmental filter override internal niche-driven relationships, resulting in tightly intertwined dynamics where taxa are affected in similar ways. In other words, our results suggest that intense disturbance may have fostered emergent cooperation and self-organised regulation, where collective resilience becomes the primary mechanism for community persistence.

The structural reorganisation of these communities is further evidenced by changes in out-degree distributions, which measure the number of taxa a single taxon ”Granger-causes”. As shown in Figure 4, out-degree frequencies vary markedly between fire regimes. During low-fire snapshots, the distribution is flatter, indicating that ecological influence is evenly distributed among many taxa. This modular, decentralised architecture enhances the robustness of the ecological network by preventing the collapse of a few critical ”hubs” from destabilising the entire system.

In contrast, high-fire periods are characterised by a distribution where a few species exert disproportionate influence over the system's trajectory. These central taxa respond faster to perturbations or act as proxies for the predictive state of the wider community, effectively becoming the drivers of synchronised recovery or decline. The coarse-grained networks (Fig 2) map these influences onto functional groups, showing that the ”Lower forest limit” and ”Afroalpine” communities in GGU, and ”Grassland” and ”Pinewoods” in BSM, act as critical nodes whose causal dependencies are radically rewired as fire regimes shift. **desarrollar un poco más en relación a la ecologia de esas comundiades**

Our study highlights a critical methodological requirement for applying causal inference to the palaeoecological record: the prioritisation of Pollen Accumulation Rates (PAR) over relative percentages. Palynological data are traditionally interpreted through percentages, which are subject to the ”closure effect”—a mathematical artifact where an increase in one taxon necessitates a decline in another to maintain a sum of 100%. In the context of Granger Causality (GC), using percentages can lead to false inferences of negative causality that reflect mathematical constraints rather than biological interactions

Perhaps the most thought-provoking result is the high volatility of community temporality. Our analysis of node turnover and interaction persistence (see ”Community temporality”) suggests that even when the same taxa are present over millennia, the coexistence patterns are not static. This indicates that dynamic rewiring is a key mechanism for ecological resilience. Communities do not simply persist by remaining unchanged; they survive by constantly reorganising their causal dependencies in response to shifting environmental filters.

This ”persistence through reorganisation” suggests that the resilience of high-altitude ecosystems is intrinsically tied to the community's capacity to shift its assembly mode. The Pyrenean and Ethiopian mountains, acting as biodiversity hotspots and glacial refugia, have demonstrated a remarkable ability to withstand abrupt climate events (e.g., the 8.2 ka event) and fluctuating fire frequencies through the Holocene. However, as anthropogenic pressures push ecosystems beyond their safe operating boundaries, the simplification of these networks—particularly the loss of niche partitioning during low-disturbance phases—may reduce the community's capacity to buffer against critical transitions.

Our findings have profound implications for managing endangered ecosystems in the Anthropocene. By tracing the baseline conditions of ecosystem complexity prior to the "Great Acceleration," we provide a quantitative framework for assessing how current disturbances might reorganise future communities. The fire-modulated dichotomy between synchrony and independence suggests that conservation policies must account for internal community architecture as much as individual species protection. Identifying key taxa in ecological networks—those that maintain connectivity or drive recovery—will be essential for designing agro-forestry measures that prevent major loss of ecosystem services. Ultimately, understanding how long-term persistence is achieved through causal flexibility offers a vital bridge between palaeo- and neo-ecological sciences, providing a more robust basis for ecological forecasting in a rapidly changing world.

[GRANGER CAUSALITY LIMITATIONS]

During the development of this work, several causal inference methodologies were considered before settling on GC. One of its main advantages lies in its very low dimensionality, especially when compared to others such as Convergent Cross Mapping and Transfer Entropy and when taking into account the dataset's limitations[22]. While the BSM and GGU datasets consist of 140 and 301 data points respectively. 140 points can comfortably be fitted with a straight line (or, depending on the amount of conditional variables, a three or four dimensional plane). However, these same 140 points cannot densely fill a four dimensional space. On one hand, CCM requires that most of these points repeatedly land near each other in a high-dimensional space to find the nearest neighbors. On the other hand, TE needs to estimate the joint probability distributions of all variables in an even higher dimensional space that is mostly empty (for example, a one-dimensional histogram that consists of ten boxes would grow to 100,000 boxes in 5 dimensions). For these reasons, GC's assumption of linearity and low dimensionality allows for robust statistical inference even with constrained temporal data.

However, these assumptions have important limitations of their own. First, GC relies on an autoregressive model that assumes the separability of variables. In this situation, the causal effect from one taxon to another is treated as linear and additive, implying that the influence of a causing species remains constant regardless of the population size of the caused species or the overall state of the ecosystem. Furthermore, while GC models causality between two taxa in both directions independently, it does not capture the feedback loops that models like CCM can take into account. Lastly, a problem that is common to most causal inference methods is the lack of stationarity, which is addressed by separating the time series into snapshots based on different fire regimes.

[DATA LIMITATIONS]

The interpretation of the inferred networks must be contextualized within the inherent limitations of paleoecological data collection. The time series are constructed using pollen counts, which serve as a proxy for plant biomass rather than a direct measurement. This introduces representational bias, as taxa possess vastly different pollen production rates and dispersal capabilities. Additionally, sedimentary data that originates from a single site may not reflect the abundance of a taxon over the entire ecosystem, and is subject to variation depending on the change in location of the species (or even the spread of its pollen).

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Author contributions statement

GGR, HS, PVR, AC, **¿OTROS?** designed the study, GGR, HS, PVR, and AC analyzed the results, GGR, PVR, AC wrote the paper. PVR and AC conceptualized the visualizations, GGR, PVR, and AC wrote the first draft. PVR and AC performed the simulations, curated the data, code, and visualizations. All authors read, reviewed, and approved the final manuscript.

Additional information

Competing interests The authors declare no competing interests.

Availability of data and materials @@@@@ DATA AVAILABILITY @@@@@

Ethical approval This article does not contain any studies with human participants performed by any of the authors.

Informed consent This article does not contain any studies with human participants performed by any of the authors.

References

1. Granger, C. W. J. Investigating Causal Relations by Econometric Models and Cross-spectral Methods. *Econometrica* **37**, 424–438, DOI: 10.2307/1912791 (1969). 1912791.
2. Politis, D. N. & Romano, J. P. The Stationary Bootstrap. *J. Am. Stat. Assoc.* **89**, 1303–1313, DOI: 10.1080/01621459.1994.10476870 (1994).
3. Latora, V., Nicosia, V. & Russo, G. *Complex Networks: Principles, Methods and Applications* (Cambridge University Press, Cambridge, 2017).
4. Newman, M. E. J. The Structure and Function of Complex Networks. *SIAM Rev.* **45**, 167–256, DOI: 10.1137/S003614450342480 (2003).
5. Wasserman, S. & Faust, K. *Social Network Analysis: Methods and Applications*. Structural Analysis in the Social Sciences (Cambridge University Press, Cambridge, 1994).
6. Sporns, O. *Networks of the Brain* (MIT Press, Cambridge, MA, USA, 2016).
7. Barabási, A.-L. The network takeover. *Nat. Phys.* **8**, 14–16, DOI: 10.1038/nphys2188 (2012).
8. Pilosof, S., Porter, M. A., Pascual, M. & Kéfi, S. The multilayer nature of ecological networks. *Nat. Ecol. & Evol.* **1**, DOI: 10.1038/s41559-017-0101 (2017).
9. Kéfi, S. *et al.* Advancing our understanding of ecological stability. *Ecol. Lett.* **22**, 1349–1356, DOI: 10.1111/ele.13340 (2019).
10. Pérez-Sanz, A. *et al.* Holocene Climate Variability, Vegetation Dynamics and Fire Regime in the Central Pyrenees: The Basa de La Mora Sequence (NE Spain). *Quat. Sci. Rev.* **73**, 149–169, DOI: 10.1016/j.quascirev.2013.05.010 (2013).
11. Gil-Romera, G. *et al.* The new Garba Guracha palynological sequence: Revision and data expansion. In *Quaternary Vegetation Dynamics*, 127–134, DOI: 10.1201/9781003162766 (CRC Press, 2021).
12. Stockmarr, J. Determination of spore concentration with an electronic particle counter. In *DGU Årbog 1972*, vol. 1972, 87–89, DOI: 10.22008/gpub/38152 (DGU, 1973).
13. Hastie, T. & Tibshirani, R. Generalized Additive Models. *Stat. Sci.* **1**, 297–318 (1986).
14. Servén, D. *et al.* pyGAM: v0.10.1. Zenodo, DOI: 10.5281/zenodo.16453423 (2025).
15. Cazelles, K., Araújo, M. B., Mouquet, N. & Gravel, D. A theory for species co-occurrence in interaction networks. *Theor. Ecol.* **9**, 39–48, DOI: 10.1007/s12080-015-0281-9 (2016).
16. Coenen, A. R. & Weitz, J. S. Limitations of Correlation-Based Inference in Complex Virus-Microbe Communities. *mSystems* **3**, 10.1128/msystems.00084–18, DOI: 10.1128/msystems.00084-18 (2018).
17. Krakovská, A. *et al.* Comparison of six methods for the detection of causality in a bivariate time series. *Phys. Rev. E* **97**, 042207, DOI: 10.1103/PhysRevE.97.042207 (2018).
18. Bernal-Wormull, J. L. *et al.* New insights into the climate of northern Iberia during the Younger Dryas and Holocene: The Mendukilo multi-speleothem record. *Quat. Sci. Rev.* **305**, 108006, DOI: 10.1016/j.quascirev.2023.108006 (2023).
19. Lahiri, S. N. *Resampling Methods for Dependent Data*. Springer Series in Statistics (Springer New York, New York, NY, 2003).
20. Gil-Romera, G. *et al.* Biomass-modulated fire dynamics during the Last Glacial–Interglacial Transition at the Central Pyrenees (Spain). *Palaeogeogr. Palaeoclimatol. Palaeoecol.* **402**, 113–124, DOI: 10.1016/j.palaeo.2014.03.015 (2014).
21. Gfeller, D. & De Los Rios, P. Spectral Coarse Graining of Complex Networks. *Phys. Rev. Lett.* **99**, 038701, DOI: 10.1103/PhysRevLett.99.038701 (2007).
22. Runge, J. *et al.* Inferring causation from time series in Earth system sciences. *Nat. Commun.* **10**, 2553, DOI: 10.1038/s41467-019-10105-3 (2019).

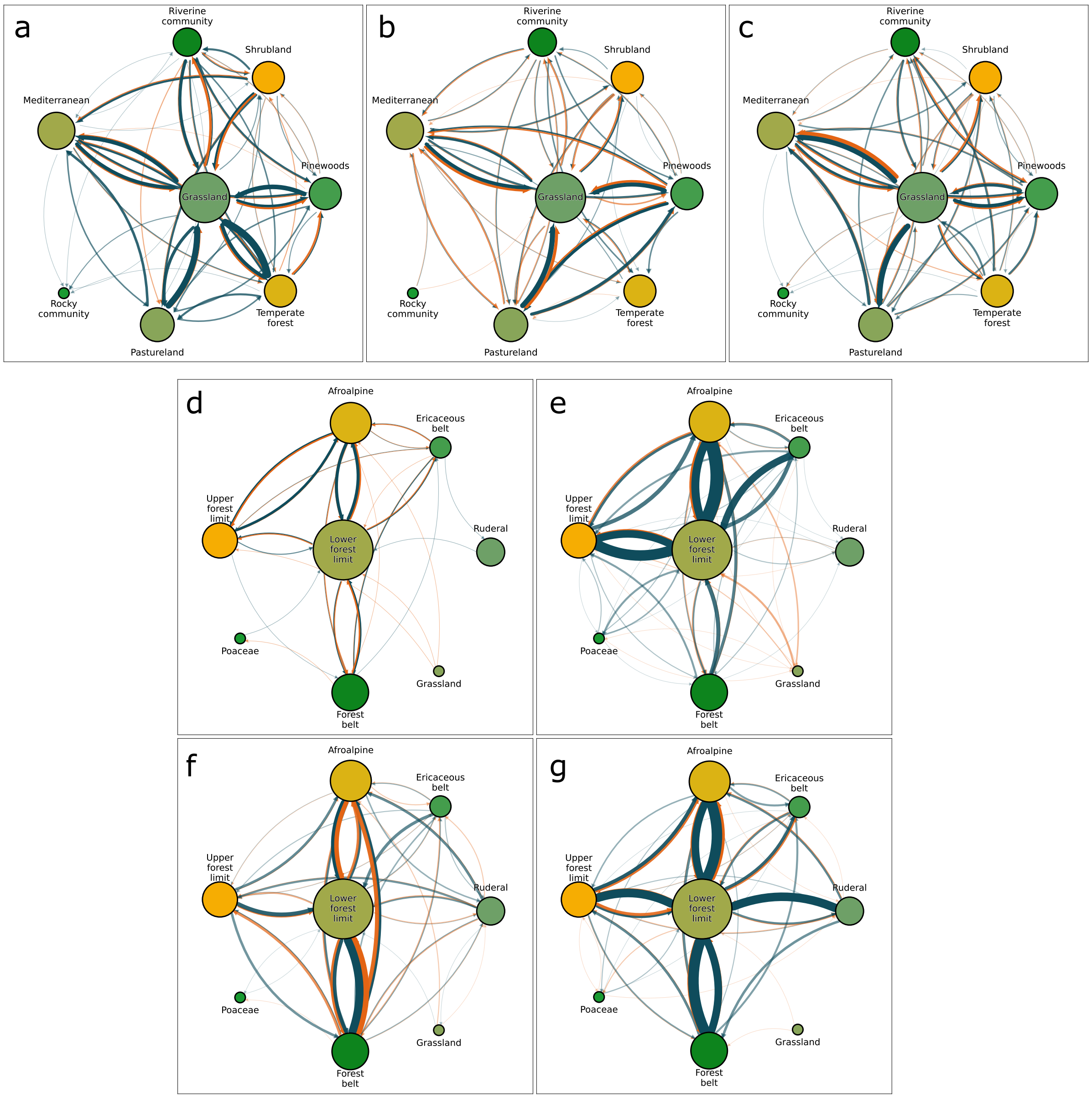

Figure 2. Causal networks of the BSM (panels a–c) and GGU (panels d–g) sites. Each node represents a functional group of taxa and a link between them encodes the amount of causal relationships existing between taxa belonging to distinct groups. The hue of the link denotes the sign of the relationship (green for positive and orange for negative), whereas its direction is given by the arrowhead. **@@ A COMPLETAR @@**.

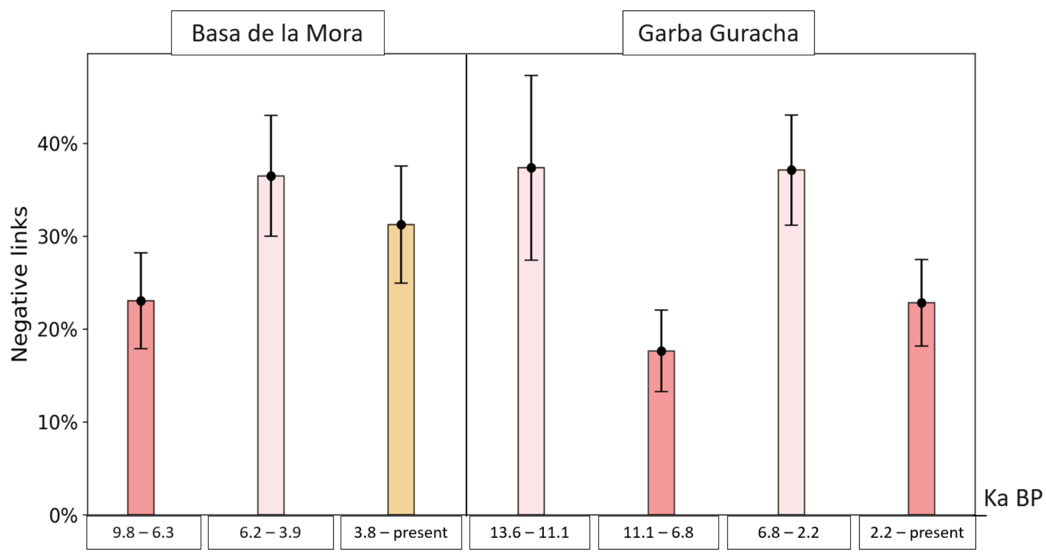

Figure 3. Percentage of negative causal links found for every snapshot in BSM (left panel) and GGU (right panel). The hue of the bar denotes the fire anomalies intensity for that snapshot, with darker shades representing higher fire intensities. @@ EXPLICAR A QUE CORRESPONDEN LAS BARRAS D'ERROR @@

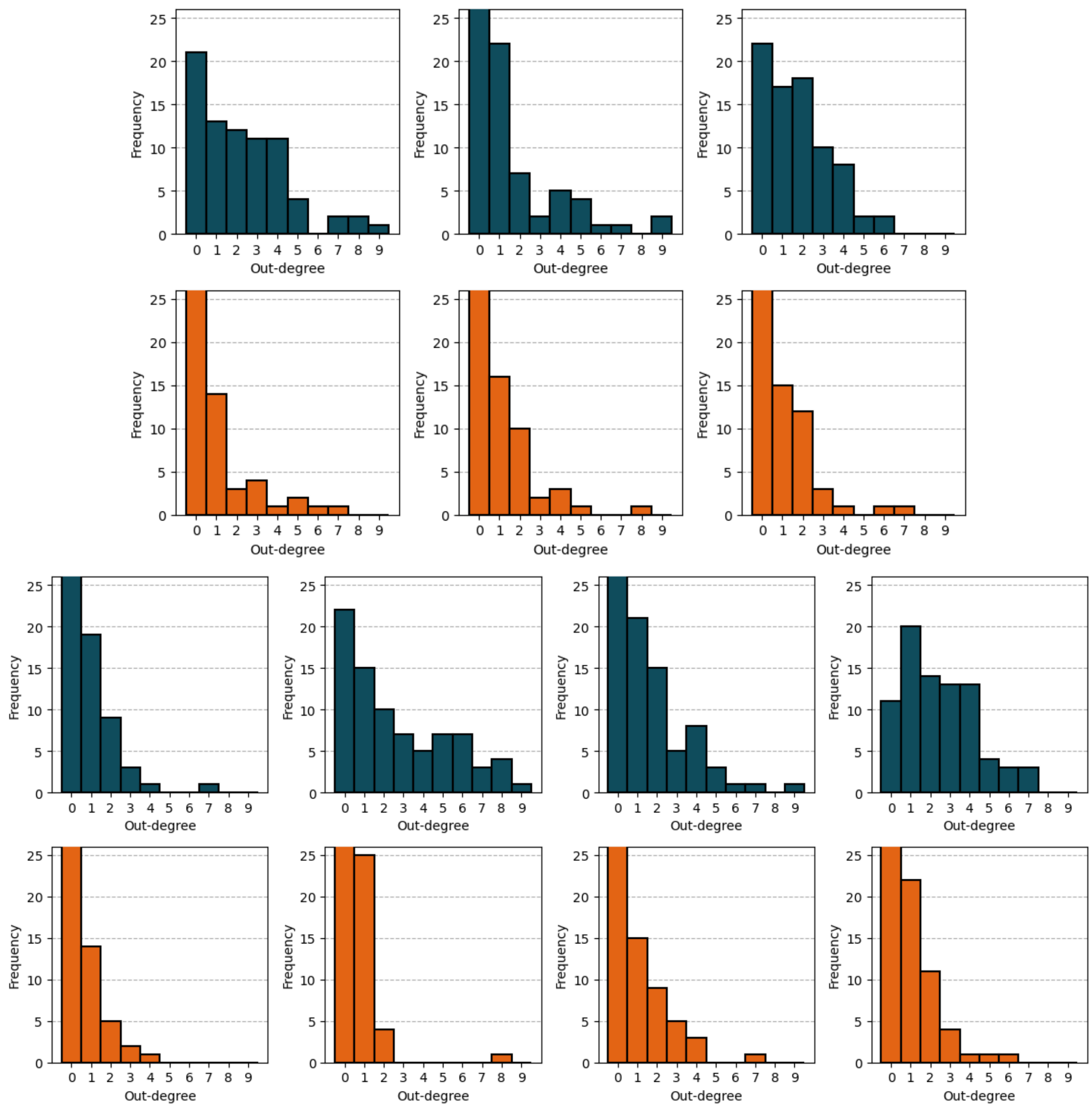

Figure 4. [NUMERADAS A-N] **a-f** Histograms from the BSM dataset, showing outgoing positive **a-c** and negative **d-f** links for each snapshot. **g-n** Corresponding histograms from the GGU dataset, representing outgoing positive **g-j** and negative **k-n** links.